Name: _____ Date: _____ Period: _____

Unit 1 Practice Quiz #2

1) Which expression is equivalent to $6^4 \times 6^{-6}$?

a) $\frac{1}{36}$ c) 6^{-10}

b) 36 d) $-\frac{1}{36}$

2) Which expression is **not** equivalent to $\frac{2^5}{2^{10}}$?

a) $\frac{1}{2^2}$ c) 2^{-5}

b) $\frac{1}{32}$ d) $\frac{1}{2^5}$

3) Re-write using positive exponents.

8^{-2}

4) Simplify the expression using **positive exponents.**

$\frac{7^3}{7^8}$

5) Simplify the expression using **positive exponents**.

$(2^4)^{-3}$

6) Simplify the expression using **positive exponents.**

$(5^3)^{-2}$

7) Simplify the expression using positive exponents.

$$3^{-8} \cdot 3^{3}$$

8) Simplify the expression using **positive exponents.**

$$2^{-9} \cdot 2^{4}$$

9) Simplify the following in exponential notation.

$$7^{8} \cdot 7^{-5}$$

10) Mia says the following expressions are equivalent.

$$\frac{4^{3}}{4^{7}} \quad \text{and} \quad 4^{-2}$$

Is Mia correct? Use the properties of exponents to explain your answer.